Solutions 1:

Question 1

(a) The energy levels E_n in a 1-D quantum well of width L, measured from the bottom of the well E_w, are given by $E_1 = \frac{n^2\pi^2\hbar^2}{2mL^2}$ where $n \geq 1$. These levels are shown in Fig. A1.1 below:

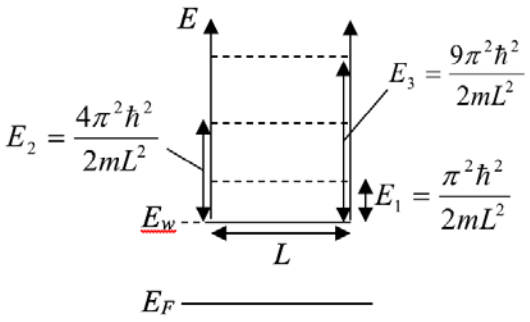


Figure A1.1

As the probability that level E_1 is occupied is 0.5%, this implies that the Fermi energy E_F lies below E_1 (Fig. A1.1). Using the Fermi-Dirac distribution:

$$f(E_1 + E_W) = \frac{1}{1 + exp\left(\frac{E_1 + E_W - E_F}{k_B T}\right)} = 0.005$$

$$\Longrightarrow 1 + exp\left(\frac{E_1 + E_W - E_F}{k_B T}\right) = 200$$

$$\Longrightarrow exp\left(\frac{E_1 + E_W - E_F}{k_B T}\right) = 199$$
$$\Longrightarrow E_1 + E_W - E_F = k_B T ln(199)$$
$$\Longrightarrow E_1 + E_W - E_F = 0.0258 \times 5.293 = 0.136\ eV$$

[3 marks]

Furthermore, $E_1 = \frac{\pi^2\hbar^2}{2mL^2} = \frac{\pi^2\left(1.055\times10^{-34}\right)^2}{2(10^{-30})(5\times10^{-9})^2} = 0.0135eV$

$E_2 = 4E_1 = 0.054eV$

$\therefore E_w - E_F = (E_1 + E_w - E_F) - E_1 = 0.136 - 0.0135 = 0.123\text{eV}$

$$\therefore f(E_2 + E_W) = \frac{1}{1 + exp\left(\frac{E_2 + E_W - E_F}{k_B T}\right)} = \frac{1}{1 + exp\left(\frac{0.054 + 0.123}{0.0258}\right)} = 0.001 = 0.1\%$$

[4 Marks]

(b) Assume the wave-function ψ is represented by a travelling wave:
$\psi = Aexp\big(i(kx - \omega t)\big)$ where wave vector $k = 2\pi/\lambda$

For a free particle in a region with potential energy V:

Total energy = kinetic energy + potential energy

$\Rightarrow \quad E = \frac{p^2}{2m} + V$ Eq. 1

where p is the momentum. Furthermore, $p = \hbar k$ *and* $E = \hbar\omega$

[Marks: 3]

Using the wave-function ψ, we see that:

$$\frac{\partial\psi}{\partial t} = -i\omega\psi \quad \text{and} \quad \frac{\partial^2\psi}{\partial x^2} = -k^2\psi$$

Substituting $p^2 = \hbar^2k^2$ and $E = \hbar\omega$ into Eq. 1, and then using the above two expressions to eliminate k and ω gives us the required equation:

$$\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = i\hbar\frac{\partial\psi}{\partial t}$$

where $\psi\psi^*$ represents the probability density.

[Marks: 4]

(d)

We scale the MOSFET such that the channel length L, width W and oxide thickness t_{ox} are reduced as follows:-

L reduces to L/K

W " " W/K

t_{ox} " " t_{ox}/K.

For doping concentration, Poisson's Eq. relates the electric fields F_x and F_y to the depletion layer charge:-

$$\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} = \frac{-eN_A}{\varepsilon_{Si}\varepsilon_0}$$ (with p-type Si 'bulk' region)

2 Marks

So, to keep F_x and F_y constant when dimensions along x and y are scaled by $1/K$, N_A (or N_D) must go up by 'K', i.e:-

N_A, N_D increased to KN_A, KN_D.

2 Marks

For voltages 'V', as $F \sim V/L$, reducing L by L/K requires that V is reduced by V/K.

Then (for gate field along 'y' axis):-

Scaled value of $F_y \sim \frac{V_g/K}{t_{ox}/K} = \frac{V_g}{t_{ox}}$ → remains constant

Scaled value of $F_x \sim \frac{V_d/K}{L/K} = \frac{V_d}{L}$ → remains constant

2 Marks

Solution 2:

a) The energy band diagrams along the line *AA'* are shown in Fig. A3.1 below:

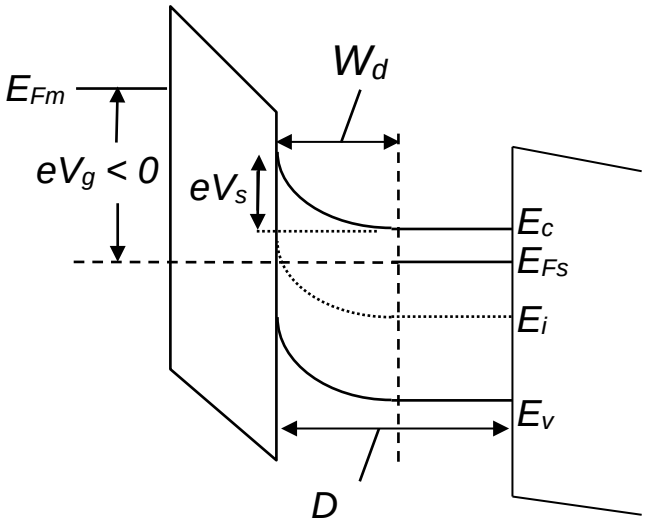


Fig. A2.1. Energy bands along AA', with the channel partially depleted. Full depletion occurs at $V_g = V_{th}$, when $W_d = D$

[5]

(b) The charge density $\rho(x)$ vs. x, along the line *AA'*, is shown in Fig. A2.2 below:

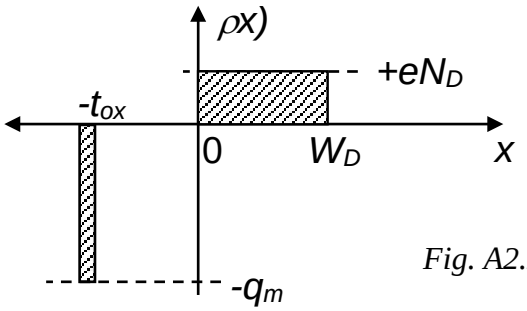


Fig. A2.2

[5]

(c) Poisson's equation, applied to the Si channel, is as follows:

$$-\frac{\partial^2 V}{\partial x^2} = \frac{\partial F}{\partial x} = \frac{\rho}{\varepsilon_{Si}\varepsilon_0} = \frac{eN_D}{\varepsilon_{Si}\varepsilon_0}$$

[3]

$\therefore$ Solving for electric field F:

$$F = \int \frac{eN_D}{\varepsilon_{Si}\varepsilon_0} dx = \frac{eN_D x}{\varepsilon_{Si}\varepsilon_0} + c$$

Boundary condition $F = 0$ at $x = W_D \Rightarrow$

$$0 = \frac{eN_D W_D}{\varepsilon_{Si}\varepsilon_0} + c \Rightarrow c = -\frac{eN_D W_D}{\varepsilon_{Si}\varepsilon_0}$$

$$\Rightarrow F = -\frac{eN_D W_D}{\varepsilon_{Si}\varepsilon_0}\left(1 - \frac{x}{W_D}\right)$$

[5]

Integrating again, for potential $V(x)$:

$$V = -\int F dx = +\int \frac{eN_D W_D}{\varepsilon_{Si}\varepsilon_0}\left(1 - \frac{x}{W_D}\right)dx = \frac{eN_D W_D}{\varepsilon_{Si}\varepsilon_0}\left(x - \frac{x^2}{2W_d}\right) + c$$

Boundary condition $V = 0$ at $x = W_D \Rightarrow$

$$c = -\frac{eN_D W_D^2}{2\varepsilon_{Si}\varepsilon_0}$$

$$\Rightarrow V = -\frac{eN_D W_D^2}{2\varepsilon_{Si}\varepsilon_0}\left(-\frac{2x}{W_D} + \left(\frac{x}{W_D}\right)^2 + 1\right)$$

$$\Rightarrow V = -\frac{eN_D W_D^2}{2\varepsilon_{Si}\varepsilon_0}\left(1 - \frac{x}{W_D}\right)^2$$

[5]

$\therefore$ At $x = 0$, the surface potential V_s is:

$$V_s = -\frac{eN_D W_D^2}{2\varepsilon_{Si}\varepsilon_0}$$

[2]

(d) The threshold voltage V_{th} corresponds to the voltage where the Si layer is completely depleted, i.e. $W_D = D$. We then have:

$$V_{th} = V_{ox} + V_s = -\frac{eN_D W_D}{C_{ox}} + V_s$$

Substituting for V_s from part (c), and using $W_D = D$ gives V_{th}:

$$V_{th} = -\frac{eN_D D}{C_{ox}} - \frac{eN_D D^2}{2\varepsilon_{Si}\varepsilon_0}$$

Here, gate capacitance $C_{ox} = \varepsilon_0\varepsilon_{ox}/t_{ox}$.

[5]

 Copyright Imperial College London

Solution 3:

a) General forms of the wave-functions ψ_1, ψ_2 and ψ_3 in the regions $x < -a/2$, $-a/2 \leq x \leq a/2$, and $x > +a/2$ respectively, are as follows:

Within the well: $\psi_2 = A\sin k_1 x + B\cos k_1 x$

[2]

Outside the well, the wave-function has the form:

$$\psi = C\exp(k_2 x) + D\exp(-k_2 x)$$

As the wave-function must be normalisable as x tends to $\pm\infty$, this implies:

$\psi_1 = C\exp(k_2 x)$ for $x < -a/2$

$\psi_3 = D\exp(-k_2 x)$ for $x > a/2$

[2 + 2]

b) The wave-vectors k_1 and k_2 within and outside the well respectively are as follows:

$$k_1 = \sqrt{\frac{2m\left(E + V_0\right)}{\hbar^2}} \quad \text{and} \quad k_2 = \sqrt{\frac{2m\left(-E\right)}{\hbar^2}}$$

where $-E > 0$. This implies that k_2 is real.

[2 + 2]

c) Applying boundary conditions to match the wave-functions, and the derivative of the wave-functions at $x = a/2$ and $x = -a/2$:

At $x = a/2$:

$$A\sin(k_1 a/2) + B\cos(k_1 a/2) = D\exp(-k_2 a/2) \qquad \text{Eq. 1}$$

$$k_1 A\cos(k_1 a/2) - k_1 B\sin(k_1 a/2) = -k_2 D\exp(-k_2 a/2) \qquad \text{Eq. 2}$$

[2]

At $x = -a/2$:

$$-A\sin(k_1 a/2)+B\cos(k_1 a/2)=C\exp(-k_2 a/2) \quad \text{Eq. 3}$$

$$k_1 A\cos(k_1 a/2)+k_1 B\sin(k_1 a/2)=k_2 C\exp(-k_2 a/2) \quad \text{Eq. 4}$$

[2]

Simplifying Eqs. 1 – 4 as follows:

Eq. 1 – Eq. 3 $\Rightarrow$ $2A\sin(k_1 a/2)=\left(D-C\right)\exp(-k_2 a/2)$ Eq. 5

Eq. 1 + Eq. 3 $\Rightarrow$ $2B\cos(k_1 a/2)=\left(D+C\right)\exp(-k_2 a/2)$ Eq. 6

[2]

Eq. 4 – Eq. 2 $\Rightarrow$ $2k_1 B\sin(k_1 a/2)=k_2\left(D+C\right)\exp(-k_2 a/2)$ Eq. 7

Eq. 4 + Eq. 2 $\Rightarrow$ $2k_1 A\cos(k_1 a/2)=-k_2\left(D-C\right)\exp(-k_2 a/2)$ Eq. 8

[2]

Equations 5 – 8 give us the relationships between the wave-vectors:

Eq. 8 / Eq. 5 $\Rightarrow$ $k_1\cot(k_1 a/2)=-k_2$ Eq. 9

Unless $C = D$ and $A = 0$

Eq. 7 / Eq. 6 $\Rightarrow$ $k_1\tan(k_1 a/2)=k_2$ Eq. 10

Unless $C = -D$ and $B = 0$

[2]

Equations 9 and 10 cannot be satisfied simultaneously, implying two types of solutions.

Finally, we may substitute $\alpha = k_1\frac{a}{2}$ and $\beta = k_2\frac{a}{2}$ into Eqs. 9 and 10 to give the form required by the question:

$\alpha\tan(\alpha)=\beta$ (Class 1 solutions)

and $-\alpha\cot(\alpha)=\beta$ (Class 2 solutions)

[2]

d) The Class 1 and 2 solutions are restricted by the following conditions:

For solutions of Class 1 to exist, we have $C = D$ and $A = 0$
For solutions of Class 2 to exist, we have $C = -D$ and $B = 0$

[2 + 2]

These restrictions give us the following forms for the wave-functions within the well:

Class 1: $A = 0 \Rightarrow \psi_2 = B\cos(k_1 x)$ Even solution

Class 2: $B = 0 \Rightarrow \psi_2 = A\sin(k_1 x)$ Odd solution

[2 + 2]

Answer, Question 1:

d) We need to calculate the second moment of area referred to the neutral axis:

$$I = \int (y - y_c)^2 dA = \int_0^h (y - y_c)^2 w dy = \int_0^h y^2 w dy - 2y_c \int_0^h y w dy + y_c^2 \int_0^h w dy$$

The last term on the right is equal to $+y_c^2 A$ where $A = (b + a)h/2$ is the area of the cross-section, while the penultimate term is equal to $-2y_c^2 A$ (this follows from the definition of y_c). We therefore just need to evaluate the third term in from the right.

The width *w* varies with *y* as $w = b - (b - a)y/h$, so:

$$\int_0^h y^2 w dy = \frac{bh^3}{3} - (b - a)\frac{h^3}{4} = \frac{(b + 3a)h^3}{12}$$

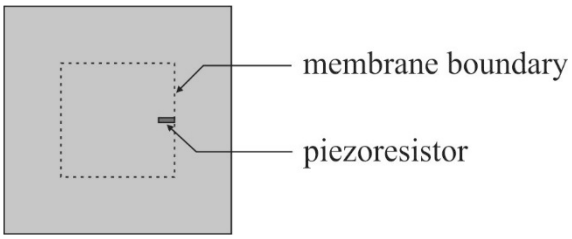
Putting the above results together:

$$I = \int_0^h y^2 w dy - y_c^2 A = \frac{(b + 3a)h^3}{12} - \frac{(b + 2a)^2}{9(b + a)^2}\frac{(b + a)h}{2}$$

which, after some rearrangement, gives the required result.

[Use of standard results for rectangle and triangle along with the parallel axis theorem is also fine.] [6]

e) The membrane stress is maximum at the midpoints of the membrane edges, so the piezoresistor should be placed at one of these points:



The piezoresistor will experience a longitudinal stress $\sigma_L = \sigma_{max}$ and a transverse stress $\sigma_T = \nu\sigma_{max}$ where ν is Poisson's ratio. Using the relations given in the *Information for Candidates*, the fractional change in resistance is:

$$\frac{\Delta R}{R} = (\pi_L + \nu\pi_T)\sigma_{max} \approx \frac{3}{20}\frac{L^2}{h^2}(1 - \nu)\pi_{44} p$$

where we have used the fact that π_{44} is much larger in magnitude than π_{11} and π_{12}. [5]

f) A given beam will buckle when the compressive stress reaches a critical value $\sigma_C = P_C/(wh)$ where w is the beam width, h is the polysilicon layer thickness and P_C is the Euler critical load. Using the equation given in the *Information for Candidates*, and putting $I = h^3w/12$ (because the beam is going to buckle out-of-plane) we obtain:

$$\sigma_C = \left(\frac{2\pi h}{L}\right)^2 E$$

With E = 170 GPa and h = 2 μm, the beam lengths given in the question correspond to stress levels of 5, 10, 15, 20, 25 and 30 MPa. The first three beams are buckled, so we know the stress is compressive and lies in the range 15 to 20 MPa. [4]

g) From the equation given in the Information for Candidates, the self-resonance frequency of the lowest order mode is:

$$f_0 = \frac{3.52}{2\pi}\sqrt{\frac{Ewh^3}{12L^3m}} \quad \text{Hz}$$

where w = 20 μm, h = 5 μm, L = 200 μm are the width, depth and length of the cantilever, $m = 4.87 \times 10^{-11}$ kg is the total mass (including the polymer layer), and E = 170 GPa. The mass is calculated as $m = wL(h\rho_{Si} + t\rho_{Po})$, with ρ_{Si} = 2330 kg/m^3, ρ_{Po} = 1050 kg/m^3 and t = 0.5 μm. Substituting these values gives an initial frequency of f_0 = 168.9 kHz.

Since $f_0 \propto m^{-0.5}$ it follows that the frequency shift due to a small change in mass will be:

$$\Delta f_0 \approx -0.5 f_0 \frac{\Delta m}{m}$$

With Δm = 1 pg, this gives $\Delta f_0 \approx -1.73$ Hz. [5]

Answer, Question 4:

a) The suspension consists of four folded flexures in parallel. Each folded flexure has a stiffness of $6EI/L^3$, so the total in-plane stiffness is $k_x = 24EI/L^3 = 2Ew^3h/L^3$. Putting E = 200 GPa, w = 10 μm, h = 10 μm, L = 450 μm we obtain k_x = 43.9 N/m. The mass of the moving element (neglecting the suspension) is m = 4 mm^2 × 10 μm × 8900 kg/m^3 = 3.56 × 10^{-7} kg. The in-plane resonance frequency is then obtained as $\omega_{0x} = \sqrt{k_x/m}$ = 11,104 rad/s corresponding to 1,767 Hz. [6]

Since the flexures are square in cross-section, we would expect $k_x = k_z$ and hence $\omega_{0x} = \omega_{0z}$. Possible causes of mismatch include:

- Fabrication errors that make the beams other than square;
- Bending of the outer spars on the suspension which will slightly lower the out-of-plane stiffness;
- Squeeze film effect which will tend to increase the out-of-plane stiffness. [2]

b) Consider the right-hand comb-drive. Each electrode gap contributes a capacitance $lh\epsilon_0/g$ where l is the overlap length. Using the virtual work method, the force contribution from each gap is $\delta F_B = \partial/\partial l(\frac{1}{2} lh\epsilon_0 V_B^2)/g = \frac{1}{2} h\epsilon_0 V_B^2/g$. The total force due to the right-hand actuator is then $F_B = \frac{1}{2} Nh\epsilon_0 V_B^2/g$ in the +x direction. By a similar argument the force due to the left-hand actuator is $F_A = -\frac{1}{2} Nh\epsilon_0 V_A^2/g$. The total electrostatic force is therefore:

$$F_x = \frac{Nh\epsilon_0}{2g}(V_B^2 - V_A^2) = \frac{Nh\epsilon_0}{2g}(V_B + V_A)(V_B - V_A) = \frac{2Nh\epsilon_0}{g} V_0^2 \cos\omega_{0x} t$$

[6]

The amplitude of the response at resonance will be $A = Q_x F_{x0}/k_x$ where F_{x0} is the amplitude of the forcing function. So, we have:

$$A = \frac{2Q_x Nh\epsilon_0 V_0^2}{k_x g} \quad \Rightarrow \quad V_0 = \sqrt{\frac{k_x g A}{2Q_x Nh\epsilon_0}}$$

Putting k_x = 43.9 N/m, g = 2 μm, A = 10 μm, Q_x = 1000, N = 98, h = 10 μm gives V_0 = 7.1 V. [4]

c) We need only the z-axis equation of motion:

$$m\ddot{v}_z + c_z\dot{v}_z + k_z v_z - 2m\Omega\dot{v}_x = F_z$$

Assuming $\omega_{0x} = \omega_{0z}$, the z-axis will be driven at resonance and the mass and spring terms will cancel out. Also, since the device is operated in open loop, F_z = 0. The EOM therefore reduces to $c_z\dot{v}_z - 2m\Omega\dot{v}_x = 0$. The relationship between the vibration amplitudes A and B in x- and z- is then:

$$B = \frac{2m\Omega}{c_z} A = \frac{2\Omega Q_z}{\omega_0} A$$

[4]

Putting Ω = 0.02 rad/s, Q_z = 100, ω_0 = 11,104 rad/s, A = 10 μm gives B = 3.6 nm. [4]

d) The sense capacitor has a value $C = A_m \epsilon_0 / d$ where A_m is the area of the proof mass. Since $C \propto d^{-1}$, for small changes in g we have $\Delta C / C \approx -\Delta d / d$. The peak-to-peak amplitude of the capacitance variation is therefore:

$$\Delta C \approx \frac{2C}{d} B = \frac{2A_m \epsilon_0}{d^2} B$$

Putting A_m = 4 mm^2, d = 2 μm and B = 3.6 nm gives $\Delta C \approx$ 64 fF. [4]

Answer, Question 5:

a) Each flexure in the suspension has a stiffness of $12EI/L^3$ where $I = h^3w/12$, so the total stiffness is $k = 48EI/L^3 = 4Eh^3w/L^3$. Putting E = 200 GPa, h = 10 μm, w = 0.5 mm, L = 1.5 mm gives k = 119 N/m. [6]

b) The total force is obtained by integrating the excess pressure over the area of the pad:

$$F_p = \int_0^{R_o} (p - p_0) 2\pi r dr = \Delta p \left\{ \pi R_i^2 + \frac{2\pi}{\ln(R_o/R_i)} \int_{R_i}^{R_o} r \ln(R_o/r) dr \right\}$$

where we have used $R_o = D_o/2$ and $R_i = D_i/2$ instead of D_o and D_i for convenience.
Now, $\int r \ln(R_o/r) dr = \frac{r^2}{2} \ln(R_o/r) + \frac{r^2}{4}$, so the second term in the curly braces becomes:

$$\frac{2\pi}{\ln(R_o/R_i)} \left[\frac{r^2}{2} \ln(R_o/r) + \frac{r^2}{4} \right]_{R_i}^{R_o} = -\pi R_i^2 + \frac{\pi(R_o^2 - R_i^2)}{2 \ln(R_o/R_i)}$$

Substituting from the second equation into the first gives the required result. [8]

c) The electrostatic force on the pad is $F_e = -A\varepsilon_0 E^2/2$ where $E = V/g$ is the electric field in the gap and $A = \pi D_o^2/4$ is the pad area. Here we are neglecting end effects and the dielectric. The only other forces acting on the pad are the spring force, $F_k = k(g_0 - g)$, and the force F_p exerted by the gas. The total force is therefore:

$$F = F_p + k(g_0 - g) - \frac{\varepsilon_0 A V^2}{2g^2} \qquad (1)$$

[6]

The conditions at the point of snap-down are $F = 0$ and $\partial F/\partial g = 0$. Differentiating the force equation gives:

$$\frac{\partial F}{\partial g} = -k + \frac{\varepsilon_0 A V^2}{g^3} \qquad (2)$$

So, when $\partial F/\partial g = 0$ we have: $\varepsilon_0 A V^2 = kg^3$ (3)

Substituting (3) into (1), and setting $F = 0$, we find that $g = \frac{2}{3}(g_0 + F_p/k)$ at the point of snap-down. Substituting this value of g into (3) gives the quoted result. [6]

c) With D_o = 3.0 mm, D_i = 1.0 mm and Δp = 1000 Pa, the equation given in part b) gives F_p = 2.86 mN.

With k = 119 N/m, D_o = 3.0 mm, g0 = 2 μm and F_p = 2.86 mN, the equation given in part c) gives V_C = 99.7 V. [4]